CMER: Crowdsourced Replication of Machine Learning
Marketing Experience and Trust
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Abstract

Machine learning (ML) increasingly shapes digital marketing, yet clicks and conversion do not establish user experience (UX), trust, or causal incrementality. This critical integrative perspective synthesizes published work on personalization, disclosure, privacy, and consumer autonomy, treating Generation Z as a heterogeneous population. We propose the Experience–Trust–Outcome (ETO) framework and the Crowdsourced Marketing Evidence and Replication (CMER) protocol. CMER defines a versioned registry, guarded status transitions, independent verification, graded evidence badges, estimands, and precommitted synthesis gates. Synthetic exemplars show how apparently similar claims can differ in denominators and attribution windows. A prospective evaluation protocol specifies reproducibility and fairness endpoints. This methodological contribution proposes auditable infrastructure; it reports no new participant outcomes or evidence that crowdsourcing improves reliability.

Keywords: machine learning; digital marketing; replication; user experience; trust; evidence registry; incrementality; Generation Z

1. Introduction and positioning

Artificial intelligence (AI) supports recommendation, predictive targeting, conversational service, and adaptive content. Reported engagement gains cannot be interpreted as causal marketing lift or user benefit without explicit assignment, denominators, attribution windows, and measures of agency and accessibility [1, 3–16]. This contribution differs from the published review by Kaponis et al. [1] by proposing a prospective verification infrastructure, not a completed systematic review or empirical trial. Natural language processing (NLP) applications such as chatbots are eligible claim families.

Practical decisions include allocating retargeting spend by randomized incremental lift rather than last-click credit; retaining human escalation when chatbots reduce contact time but fail to resolve cases; and evaluating personalization when conversion improves while consent comprehension deteriorates.

2. Questions, scope, and specialist audiences

RQ1: Which claims are reproducible estimands? RQ2: How do estimates vary across designs and contexts? RQ3: Does structured replication capture null and negative results? RQ4: Which safeguards protect confidential data? RQ5: Which UX differences persist across adult age groups? Intended expertise: marketing measurement, causal inference, human-computer interaction, privacy engineering, research ethics, and reproducibility engineering.

Claim boundary: CMER does not demonstrate a causal marketing effect, generational preference, successful external replication, public repository deposit, or completed human-subject pilot. Its protocol complements but does not replace preregistration, institutional review, systematic-review reporting, or accessibility evaluation.

3. Registry and legal state machine

The machine-readable schema and guard dictionary are provided as CMER_registry_schema.json and CMER_guard_dictionary.json. Each claim record captures provenance, intervention, comparator, outcome, estimand, denominator, attribution window, assignment, risk of bias, verification badge, SHA-256, and immutable transition history. Only the following guarded transitions are legal:

G01: proposed to underway. Ethics, COI, eligibility, estimand frozen.

G02: underway to results_deposited. Packet complete; signed hashes, denominator and window.

G03: results_deposited to screened. Two independent screeners and adjudication.

G04: screened to independently_checked. V1-V7 pass; independent rerun; Gold/Silver.

G05: screened to not_independently_verifiable. Reproducibility missing after remediation.

G06: independently_checked to synthesis_eligible. Gold/Silver; non-high RoB; harmonization assessed.

G07: ANY_EXCEPT_WITHDRAWN to withdrawn. Signed withdrawal/fraud decision; appeal recorded.

G08: not_independently_verifiable to screened. New version; renewed dual screening.

Figure 1 is supplied as a separate vector PDF with accompanying alt text. A JSON Schema validates record fields; a separate guard validator checks transition semantics. A valid schema alone does not prove scientific verification.

4. Methods: packets, screening, verification, synthesis

4.1 Contributor independence and threat control

Contributors sign records through institutional identity or cryptographic signing; institutional claims require two independent reviewers. Repository commits, source identifiers, signed artifact hashes, and conflicts of interest are logged. Fabrication or duplicate uploads trigger content fingerprint review; spoofing triggers identity challenge; selective reporting triggers preregistration comparison; denominator manipulation triggers raw-unit reconciliation; coordinated influence triggers reviewer-network and duplicate-pattern audit. Sanctions escalate from correction request, to suspension with public reason code, to permanent exclusion following documented adjudication; appeals are independently reviewed within 30 days. No reviewer may assess their own contribution.

4.2 Claim sampling and bias assessment

A prospective search would cover Scopus, Web of Science, ACM Digital Library, and Business Source Complete (2015–2026), with exact executable database queries frozen before use. Dual screeners record include/exclude/uncertain and reason codes. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [2] informs the discovery flow, without redefining this Perspectives paper as a completed systematic review. Randomized trials use the revised Cochrane risk-of-bias tool (RoB 2) [19]; nonrandomized intervention studies use Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) [20]. Twenty-record calibration targets Cohen kappa at least 0.70; actual calibration remains to be performed.

4.3 Replication packet and Outcome Data Dictionary

Required fields include registered question, study setting and dates, audience and sampling frame, intervention and comparator, assignment mechanism, estimand (ITT, ATE, ATT, incremental lift, or associational difference), primary outcome ID, observation start/end, attribution window (1, 7, 14, 30, 90 or custom days), numerator where relevant, positive denominator and unit, sample size, exclusions, missing-data plan, model, point estimate and uncertainty, preregistration/deviations, consent and ethics, data access, code, environment, funding and conflicts. The filled synthetic cover sheet, outcome dictionary and coder handbook are separate supplements. The dictionary specifies instruments, scales, timing, denominators and invalid proxies; seven coder anchors address common disambiguations.

4.4 Verification and badges

V1: signed identity; V2: source identifier; V3: preregistration/deviations; V4: denominator, assignment and window; V5: code and README; V6: locked environment/enclave ticket and checksums; V7: independent recomputation of estimate and interval within prespecified tolerance. Every item must pass before a verified status. Gold requires independently reproducible code and data; Silver requires independently attested, checksum-logged enclave rerun; Bronze is

documentation only. Bronze and Not Independently Verifiable (N.I.V.) records remain map-only, not pooled. Synthetic demonstration outputs must not be labeled Gold or Silver.

4.5 Harmonization and synthesis

H(population, treatment, outcome, estimand, window) = 1 only when all five gates pass: adult eligibility and at least 80% age-band overlap; same intervention/comparator and disclosure policy; identical outcome construct and denominator unit; same estimand and assignment class; and window difference no more than 1 day for windows up to 7 days, 3 days for 8–30 days, or 10% beyond 30 days. Pooling requires H=1, at least three independent Gold/Silver estimates, no high risk of bias, and I-squared at most 50% without qualitative incompatibility. Otherwise use structured narrative or evidence-map mode; report tau-squared, exclusions, effect intervals, and independent reproduction status. These are

proposed conservative design thresholds, not empirically validated constants.

Example: 40 purchases among 1,000 randomized users within seven days and 60 purchases per 1,200 impressions within 30 days cannot be pooled: the denominator unit and window differ. Do retain original denominators; do align from permissible raw data; do report null outcomes. Do not treat impressions as unique users; do not proportionally rescale 30-day conversion to seven days; do not pool ITT with observational last-click association.

4.6 Adult Generation Z UX substudy
Prospective strata are ages 18–27, 28–44 and 45–64. Preregister quotas, randomized or AB/BA counterbalanced assignment, device/accessibility strata, exclusions and item wording. Primary outcomes include task completion, time, errors and System Usability Scale (SUS) [17]; secondary measures include UMUX-Lite [18] and the 12-item Trust in Automated Systems Scale developed by Jian, Bisantz, and Drury [22] (seven-point response format); transparency comprehension is measured separately with preregistered items, not misrepresented as a validated scale. Audit to Web Content Accessibility Guidelines
(WCAG) 2.2 Level AA [21]. Illustrative power calculation for standardized d=0.35, two-sided alpha=0.05, 80% power gives about 128 per arm before attrition; inflate to 151 per arm for 15% attrition. No study has been executed.

5. Ethics, privacy and limitations
Use minimized metadata, signed code and synthetic examples by default. Proprietary data require lawful access, non-vendor attestation, controlled enclave execution and logged hashes. Human-subject research requires applicable consent and review before recruitment. Vendor-funded studies cannot self-verify; correction, appeal, anti-retaliation, and jurisdiction-specific retention rules must be documented. Recruitment bias is mitigated by disclosed sampling frames; outcome inconsistency by dictionary validation; fraud by immutable audit artifacts; confidentiality by enclaves; selective reporting by preregistration. Operational metrics include packet completeness, independent rerun success, reviewer agreement, time to verification, and null-result capture.

6. Illustrative records and reproducibility
Two synthetic records and a completed cover sheet accompany this manuscript. They reference published research topics for illustration but their invented counts and effect estimates are not findings of those publications. The repository-ready minimal reproducible example (MRE) generates a synthetic two-arm intent-to-treat difference and checksum. Local execution and schema validation do not substitute for an externally hosted public repository, DOI, or independent third-party verification.

7. Conclusion
CMER offers a prospective, auditable path from marketing claims to independently checked evidence. Separating causal incrementality from attribution and experiential outcomes from click counts can improve the quality of marketing decisions. Its scientific value remains conditional on actual adoption, independently reproduced replications, and transparent reporting of failures.

Declarations

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Ethics and consent: This conceptual and methodological manuscript reports no newly recruited human participants or newly collected participant data. The accompanying reproducibility example uses synthetic data only; no human-subject approval or consent was required for that example. Any future empirical research must undergo applicable ethics review and obtain required consent before recruitment or data collection.

Data and code availability: The schema, guard dictionary, illustrative records, cover sheet, dictionary, handbook, validator, and synthetic minimal reproducible example are prepared for publication. Public GitHub repository URL: [PENDING PUBLICATION]. Persistent archive DOI: [PENDING ZENODO RELEASE]. No public deposit is claimed at this stage.
AI assistance: Generative AI assisted with drafting and formatting; the author remains responsible for factual accuracy, references, and final submission.

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